

Title: Getting rid of zeroes: The morphosyntax of deadjectival ‘A-verbs’ in Italian

Short title: Getting rid of zeroes

Abstract

This paper is a contribution to the ongoing debate on employing zero morphology in theoretical modelling (cf. Dahl & Fábregas 2018, Iordăchioaia & Melloni 2023). As an empirical case, we focus on Italian deadjectival ‘A-verbs’ that are either derived through overt suffixation (e.g. *solid-ific-are* ‘solidify’) or by converting the category of the base without any additional morphological marking (e.g. *ferm-∅-are* ‘stop’). Our proposal is that the need for zero derivational suffixes can be eradicated if one adopts the Nanosyntactic model of (phrasal) lexicalisation (cf. Starke 2009, Caha et al. 2024 amongst others), and that the contrast between the two types of ‘A-verbs’ can be explained away as an effect of varying sizes of lexical items rather than the absence of overt affixes. In addition, we show that our no-zero account can be extended towards the inflectional domain in both groups of verbs as well, ultimately eliminating the need for zero morphology altogether.

Keywords: deadjectival verbs, suffixation, zero derivation, Nanosyntax, phrasal lexicalisation

1 Introduction

In Italian, verbs can be derived from adjectives in three ways. The first is zero-derivation. This is a process which assigns an existing word (A) to a new word class (V) without the addition of any derivational markers to change its form. An example is given in (1a) where the adjectival

root *ferm-* is converted into a verb without any further alterations.¹ The second way is through suffixation, a process which is achieved by combining the adjectival root with a particular derivational marker. This is illustrated in (1b), where the root *solid-* requires the suffix *-ific* to become verbal. And finally, the third way is parasynthesis, a process which involves both prefixation and zero-derivation. This is shown in (1c), where the root *cald-* seems to convert like *ferm-* does but still requires the presence of the prefix *-s* to become a legitimate verb.

- (1) a. *ferm-o* → *ferm-are*
- b. *solid-o* → *solid-ific-are*
- c. *cald-o* → *s-cald-are*

In this paper, we focus on the deadjectival verbs that are formed by means of the first strategies, i.e. zero and overt suffixation, and that belong to the first conjugation class, i.e. verbs such as *fermare* and *solidificare*, that are characterised by the vowel *-a* (henceforth A-verbs). These verbs are of interest to us due to the fact that they share a number of properties, despite being created by different word formation processes. Most notably in this regard is the fact that zero-

¹ We wish to acknowledge that the status of the theme vowel is a controversial issue. Some have argued that it is a derivational marker (cf. Kayne 2016, Fábregas 2017), while others uphold the view that it is an indicator of the conjugation class a base belongs to (cf. Serianni 1989, Dardano & Trifone 2010, Dardano 2017, Van der Spuy 2020) or serves as a purely ‘ornamental’ element (cf. Oltra-Massuet 1999, Pomino & Remberger 2022). As we discuss in Section 6, we adopt neither stance but rather follow the idea presented in Fábregas (2022) that the theme vowel connects the lexical event with time and world parameters. Given this, we still consider cases such as *ferm-are* in (1) to be examples of ‘zero-derivation’.

derivation and suffixation both obtain verbs that convey meanings on the causative-inchoative continuum (which for the most part correlates with a transitive/intransitive distinction in the argument structure). As is illustrated in (2), the verb *ferma* ‘stop.3SG’ shows ambiguity between an inchoative (change-of-state), (2a), and a causative reading, (2b).²

- (2) a. Il treno (si) **ferma** in ogni stazione.
The train REFL stops in every station
‘The train stops at every station.’
- b. Il controllore **ferma** il treno.
The conductor stops the train
‘The conductor stops the train.’

The same ambiguity can be found with verbs that require an overt suffix. Consider the contrast between the inchoative use of the verb *solidifica* ‘solidify.3SG’ in (3a) and the causative use in (3b).

- (3) a. La vernice (si) **solidifica**.
The paint REFL solidifies

² The ambiguity in (2) seems partially artificial in the sense that it seems to rely on the presence or absence of the reflexive marker *si*. We will briefly come back to this matter in Section 2.2. Regarding possible semantic distinctions, our informants indicate that pragmatically speaking *Il treno ferma in ogni stazione* is slightly more common in train schedules and official announcements, whereas *Il treno si ferma in ogni stazione* tends to be preferred in more informal or narrative contexts. However, in every-day speech, there is a tendency to alternate rather freely between both constructions.

‘The paint solidifies.’

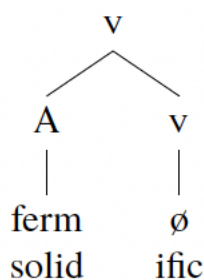
b. La vernice protettiva **solidifica** le tue creazioni.

The paint protective solidifies the your creations

‘The protective paint solidifies your creations.’

Based on these observations, it has been argued that these verbs must share the same underlying morphosyntactic structure, with surface variations arising as an effect of arbitrary phonological representations, such as the verbaliser being phonologically overt or not (e.g. Scalise 1984, Thornton 1990, Grossmann 2004). To maintain the idea of structural uniformity, though, these traditional accounts were compelled to posit the existence of a zero suffix for verbs that do not require an overt verbaliser. As depicted in (4), the advantage of postulating such a suffix is that it allows for a straightforward way to map identical morphosyntactic and semantic functions onto differently derived exponents. That is, the verbs in (2) and (3) may be derived via different processes but they remain both causative-transitive verbs.

(4)



However, there are many reasons to be sceptical about the legitimacy of zero morphemes (cf. Anderson 1992, Plag 2003, Borer 2013, Starke 2014, Dahl & Fábregas 2018 amongst others). One major issue that has been brought up is how to distinguish between the presence of a

presumably contentful zero affix and the lack of a morpheme (cf. Iordăchioaia & Melloni 2023). It has been suggested that a plausible argument for postulating a zero morpheme may be the existence of an overt one with a similar function (cf. Sanders 1998), but many authors have been critical of this view as well, arguing that zero morphemes may differ too much semantically from their apparent overt counterparts (cf. Plag 1999, Lieber 2004).

The purpose of this paper is to contribute to the ongoing debate on the use of zero morphology by providing an alternative analysis of Italian deadjectival A-verbs, one in which zero morphemes are rendered unnecessary. Our core argument is that the distinction between different derivational verb classes can be accounted for by means of cumulative exponence, more specifically, the type of exponence that is referred to as ‘phrasal lexicalisation’ in Nanosyntax. In the case of zero-derivation, we are dealing with verbal roots which are able to lexicalise all the pertinent syntactico-semantic components, thus giving the appearance of the presence of a zero marking (as mentioned before, we discuss the contribution of *si* in Section 2.2.). Conversely, in the case of suffixed verbs, a suffix is required to lexicalise some of the verbalising components. By extension, we argue that this analysis can be carried through for the whole of the present tense conjugation of both types of verbs, eliminating the need for zero suffixes altogether. The analysis is couched in Nanosyntax (cf. Starke 2009, Caha et al. 2024 amongst others), and builds on previous work on Dutch, English and Czech derived verbs in this framework (cf. Vanden Wyngaerd et al. 2022, Caha et al. 2023).

The organisation of the paper is as follows: In Section 2, we present the Italian data in more detail and address the morphosyntactic and semantic properties shared by verbs obtained through zero-derivation and suffixation. In Section 3, we revisit some of the key arguments against positing zero morphemes in the context of verbal derivation and inflection. In Section 4, we discuss the verbal structure we adopt for our proposal and introduce a few general notions

about Nanosyntax and its model of lexicalisation. Essentially, we assume that there is no single, verbalising projection, but that the verb has a complex internal structure (cf. Ramchand 2008), and that this complex structure is lexicalised in different ways, depending on whether the verb takes an overt suffix or not. In Section 5, we show the worked-out derivations of zero-derived and suffixed verbs, and show that the contrast between the two verb types arises as an effect of varying (lexical) root sizes, which correlates with whether we find additional affixes on the verb or not. In Section 6, we discuss how this proposal can easily be extended towards the rest of the present tense paradigm, and how phrasal lexicalisation can also deal with potential zeroes in the inflectional domain. Section 7 concludes and provides some avenues for future research.

2 The data

In this section, we zoom in on the two types of deadjectival A-verbs under scrutiny, i.e. the ones derived through zero-derivation or suffixation. In Section 2.1., we provide a more comprehensive overview of each word formation process and of the main verbalisers available in Italian, and in Section 2.2. we dig deeper into the shared morphosyntactic and semantic properties that have led to a unified structural analysis.

2.1 (C)overt verbalisers in Italian

As was mentioned before, Italian allows for deadjectival verbs that are not derived by any overt suffixes. As the examples in (5) show, the adjectival bases combine with the theme vowel *-a* which indicates that they belong to the first conjugation class.

(5)

SUFFIX	BASE	GLOSS	VERB	GLOSS
∅	attiv-o	‘active’	attiv-a-re/(si)	‘activate’
∅	ferm-o	‘still’	ferm-a-re/(si)	‘stop’
∅	san-o	‘healthy’	san-a-re	‘heal/cure’

Although the majority of deadjectival verbs in general are part of the first conjugation, there are also a few examples that belong to the third conjugation, ending in *-ire*. Consider, for instance, the verb *chiar-∅-ire* ‘clarify’, derived from the adjective *chiar-o* ‘clear’, or *svelt-∅-ire* ‘make quicker’, derived from the adjective *svelt-o* ‘quick’. However, this is again not productive, except when new verbs enter the system via parasynthesis. An example of this would be the verb *in-giall-∅-ire* ‘turn yellow’ (cf. Iacobini 2004). There are also a handful of verbs that allow for two distinct, inflectional patterns, despite generally not differing in meaning and being (partially) synonymous (e.g. *scuro* ‘dark’ > *scurire* / *scurare* ‘darken’, Thornton 2004: 534).

Broadly speaking, zero-derivation is a highly productive process in contemporary Italian, but only when it comes to denominal derivations (cf. Iacobini & Thornton 1992). Only about 12% of the verbs obtained through zero-derivation are derived from an adjectival base (cf. Thornton 2004: 534). Adjective-verb conversion can thus be considered as a rather minor verb formation process in Italian.

The most productive way to derive verbs is through overt suffixation. In contemporary Italian, there are many different verbalisers. The suffix *-izz* is by far the most productive one across all registers, quickly followed by *-eggi* (cf. Iacobini & de Rosa, accepted). The suffix *-ific* on the other hand is seldom used to form new verbs and is only preserved in a few cases

(cf. Grossmann 2004). Similarly, the suffixes *-ic* and *-it* only occur with some morphologically and semantically transparent verbs, but are no longer productive (cf. Grossmann 2004). Examples with each suffix are shown in (6).

(6)	SUFFIX	BASE	GLOSS	VERB	GLOSS
	-izz	stabil-e	‘stable’	stabil-izz-a-re/(si)	‘stabilise’
	-eggi	bianc-o	‘white’	bianch-eggi-a-re/(si)	‘whiten’
	-ific	intens-o	‘intense’	intens-ific-a-re/(si)	‘intensify’
	-ic	zopp-o	‘crippled’	zopp-ic-a-re	‘limp’
	-it	facil-e	‘easy’	facil-it-a-re/(si)	‘simplify’

Overall, despite the fact that there are some restrictions on the possible combinations of adjectives and suffixes, it remains largely unpredictable which adjectives combine with which suffixes, an observation which has also been made for Dutch, English and Czech derived verbs (cf. Vanden Wyngaerd 2022, Caha et al. 2023).

First, many Italian deadjectival verbs are remnants of their Latin predecessors and maintained the same suffix over time. For instance, while examining the diachronic development of the two most common suffixes *-izz* and *-eggi*, Tronci (2019) shows that Old Italian (and most likely also Late Latin) verbs ending in *-izzare* are almost exclusively loaned from Classical Latin (e.g. Lat. *sol(l)emnizare* > It. *solennizzare* ‘solemnise’), whereas the ending *-eggiare* was used for genuinely novel Italian formations (e.g. *verdeggiare* ‘turn green’). Regardless of this pattern, Tronci is still obliged to conclude that the appearance of each suffix is strongly connected with lexical idiosyncrasy.

Second, there are also some phonological restrictions at play (cf. Peperkamp 1995). Consider, for instance, the suffix *-izz*. Normally, it attaches to adjectives ending in the vowel

/e/ (with subsequent truncation of the vowel). These can be non-derived (e.g. *fragile* ‘fragile’ > *fragilizzare* ‘make fragile’) as well as already derived adjectives (e.g. *normale* ‘normal’ from *norma* ‘norm’ > *normalizzare* ‘normalise’). However, there are also exceptions, such as *verde* ‘green’³. Even though, this adjective is expected to follow the same phonological pattern, it only likes the suffix *-eggi*. In a similar vein as the diachronic pattern observed by Tronci, phonological conditions may explain why a form as **intensizzare* does not exist (i.e. the adjective *intens-o* ends in a vowel /o/), but additional factors seem to be involved as well, as some adjectives ending in /o/ do permit the condition. Take for instance, the adjective *massim-o* ‘maximum’, from which in scientific disciplines such as mathematics, the pair *massimare* and *massimizzare* ‘make a function and take on its maximum value’ is successfully derived (cf. Grossmann 2004: 461). Phonology alone cannot explain why these doublets exist. The same holds true when trying to tease apart the uses of other suffixes. Many Italian adjectives end in /o/, but there is no phonological rule which can explain why the verb is *biancheggiare* ‘whiten’ and not **bianchificare*, or why there is a verb *zoppicare* ‘limp’, but not **zoppitare*.

Finally, a last point that shows that it is unpredictable which suffix verbalises which adjective is that in several cases an adjective is compatible with multiple different suffixes, resulting in a number of doublets which can be used synonymously (e.g. *nereggiare* vs. *nericare* ‘blacken’, *zoppeggiare* vs. *zoppicare* ‘limp’), or partially synonymously (e.g. *attivare*

³ An anonymous reviewer wondered about whether the /e/ in *verde* is epenthetic, as opposed to that of *normale*. It is not. As has been discussed in, for instance, Maiden (1995), the vast majority of Italian nouns and adjectives derive from Latin accusative forms, and Italian *verde* traces back to Latin *viridis* ‘green’, whose accusative form corresponded to *viridem*. With the loss of the final *-m*, the form underlying Italian *verde* became *viride*. Thus, the /e/ is inherited, while phonological changes primarily affected the root.

vs. *attivizzare* ‘activate’, *massimare* vs. *massimizzare* ‘maximise’) (cf. Grossmann 2004: 461 for more examples of such doublets).⁴

To sum up, so far we have briefly discussed the processes of zero-derivation and suffixation in Italian, the variability of available suffixes, and the (un)predictability of their match-ups with different types of adjectives. In what follows, we will not pay further attention to specific properties of the adjective itself but focus solely on the (lack of) derivational morphology.

2.2 Causative/inchoative alternation

Let us now examine some of the main semantic and syntactic properties of the deadjectival verbs. As mentioned before, these verbs all express meanings on the causative-inchoative continuum and allow for a double reading, regardless of whether they are derived by an overt verbaliser or not. In most cases, the distinction seems to depend on whether the construction is formally marked by means of the reflexive marker *si* – in which case it is inchoative – or not. This is illustrated in (7) for the zero-derived verb *attivare* ‘activate’ and in (8) for the suffixed verb *stabilizzare* ‘stabilise’.

- (7) a. Paolo attiva il Wi-Fi.
Paolo activates the wifi.
‘Paolo activates the wifi.’

⁴ For reasons of space, we do not address formal implementations of the allomorphy of the derivational suffixes, such as between *nereggiare* and *nericare*, but we refer the reader to De Clercq & Vanden Wyngaerd (2019) for some ideas on how to approach this issue in Nanosyntax.

- b. Il Wi-Fi si attiva automaticamente.
The wifi REFL activates automatically.
'The wifi activates automatically.'
- (8) a. La nuova legge ha stabilizzato la situazione.
The new law has stabilised the situation
'The new law has stabilised the situation.'
- b. La situazione si è stabilizzata.
The situation REFL is stabilised
'The situation has stabilised.'

Importantly, however, there are also several verbs that may express both interpretations with or without the reflexive marker and therefore do not follow the strict causative/non-reflexive vs. inchoative/reflexive pattern (cf. Grossmann 2004). An example of such a verb would be *solidificare* 'solidify', shown in (9).

- (9) a. Il freddo solidifica l'acqua.
The cold solidifies the-water
'The cold solidifies the water.'
- b. L'acqua (si) solidifica.
The-water (refl) solidifies
'The water solidifies.'

Much like the examples in (7a) and (8a), (9a) only allows for a causative reading, with the external causer (*il freddo*) in the subject position and the undergoer of the causation in the object position (*l'acqua*). The example in (9b), on the other hand, does not present an external

causer, and irrespective of the presence of the *si*-marker, yields an inchoative interpretation. This can be confirmed by applying a number of diagnostic tests (cf. Cennamo 2012). One of those is the modification of the final state of the event by means of a completive adverb like *completamente* ‘completely’, as in (10).

- (10) a. #L’acqua solidifica completamente.
 The-water solidifies completely
 ‘The water solidifies completely.’
- b. #L’acqua si solidifica completamente.
 The-water REFL solidifies completely
 ‘The water solidifies completely.’

As can be seen in (10a) and (10b), the same verb *solidifica* denotes an atelic process, which by many speakers is perceived as semantically marked with an adverb adding a resultative meaning to the ongoing event.⁵ This holds true for both cases, with or without *si*, which suggests that the inchoativity or atelicity may not be an immediate effect of the reflexive marker.

⁵ It should be noted that verbs such as *solidificare* are not fully atelic, but function in a similar way to degree achievement verbs (e.g. *diminuire* ‘decrease’). As Cennamo (2012: 399) points out, these types of verbs denote processes that may unfold in multiple stages, including a potential final stage, which raises the question of whether they should not be compatible with resultative adverbs such as *completamente* (e.g. the water solidified completely to the point that it turned into ice). However, the final stage for such verbs seems to be less well-defined as for real telic verbs or gradual completion verbs, so while they are not inherently incompatible with resultative components, our informants perceive the combination of *solidificare* and *completamente* as marked and somewhat unnatural.

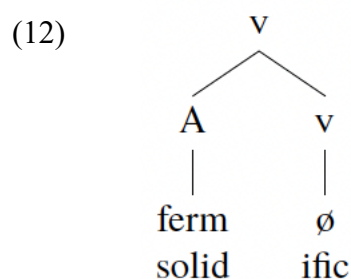
Another test that shows that we are dealing with a gradual process and not a result state is by negating the final end-state by means of a durative construction. This is infelicitous with inherently telic predicates but is possible with atelic ones. In (11a) and (11b), both sentences are acceptable. If *si* was a marker of inchoativity then its presence would thus be expected to be obligatory. Vice versa, if *si* turns an event into an atelic one, then (11a) would be expected to be ungrammatical. This again suggests that *solidificare* is atelic out of its own accord.

- (11) a. L'acqua solidificò (per un ora) ma non è solidificata.
 The-water solidified for an hour but not is solidified
 'The water solidified (for an hour) but it is not solidified.'
- b. L'acqua si solidificò (per un ora) ma non si è solidificata.
 The-water REFL solidified for an hour but not REFL is solidified
 'The water solidified (for an hour) but it is not solidified.'

Given such alternating constructions, the correlation between inchoativity and reflexivity is not self-evident. That is, the reflexive marker *si* should not be automatically presumed to be the element which is responsible for the inchoative reading of the verb. Instead, we assume it is much more plausible that inchoativity is already encoded in the verb itself. We elaborate on this encoding in Section 4.⁶

⁶ The function of *si* has been the subject of much previous work (cf. Folli 2002, Manente 2008, Schäfer 2008, Cennamo & Jezek 2011, Cennamo 2012, 2021 amongst others). A fully-fledged analysis of *si*-inchoatives would go beyond the scope of the present paper, but what we assume for now is that *si* may encode a kind of completive aspectual feature (cf. Cinque 1999: 83) which expresses the possibility of the completion of all internal stages up to the attainment of the natural endpoint of an event (cf. Caha et al. 2023 who argue that the Czech reflexive clitic *se* may spell out an aspectual feature). As such, it is not involved in the verbalisation of any core verbal subevents.

Turning back now to the main point of this section, we could conclude from the verbs in (7-9) that both zero-derived and suffixed verbs can receive a double reading. This means that they must semantically and syntactically carry properties pertaining to causativity and inchoativity. The traditional way this has been accounted for is by assuming that the information is contained within a single verbalising projection *v*, which is then realised by means of an overt verbalising suffix, or by analogy, a zero suffix, as depicted in (4) and repeated here in (12).



In our view, this is insufficient. In Section 3, we first revisit some key arguments as to why employing zero morphology in derivational processes (and by extension in inflectional ones) is undesirable. In Section 4, we then argue for a more fine-grained decomposition of the verbal structure that is better suited to account for the alternations we find.

3 No zero, no cry

Zero morphemes, i.e., morphological units which lack phonological information but nonetheless encode syntactic and/or semantic content, have been a hot topic over the past few decades (cf. Iordăchioaia & Melloni 2023 for a recent overview of the discussion). Proponents of zero see themselves especially confirmed when it comes to morphological phenomena such as inflectional syncretisms and affixless category changes.

Italian, for instance, is a language that typically has dedicated morphology to express singularity and plurality (together with gender) on nouns, (13a-b). However, in some cases number is not overtly realised, creating syncretism between the singular and plural form, (13c). To maintain a one-to-one mapping between form and function where such a semantic contrast arises without a corresponding phonological exponent, zero morphemes would have to be employed.⁷

- (13) a. la cas-**a** → le cas-**e**
 DET house-SG DET house-PL
- b. il ragazz-**o** → i raggaz-**i**
 DET boy-SG DET boy-PL
- c. la città-**∅** → le città-**∅**
 DET city-SG DET city-PL

Zero morphemes are also argued to be responsible for the derivation of nominalised infinitives (cf. Iacobini 2004). That is, when a verb like *mangiare* ‘to eat’, (14), becomes noun-like, the category change is attributed to a zero suffix.

- (14) a. Paolo deve mangiare presto.
 Paolo must eat-V soon
 ‘Paolo has to eat soon.’
- b. Il suo mangiare-**∅** presto mi irrita.

⁷ We follow the idea developed in, amongst others, Caha (2023) that a Number Phrase consists minimally of a default number feature, which is typically expounded by a singular form, and that plurals are built from this by the addition of a dedicated plural feature. In Italian, these features cluster consistently with gender.

The his eat-N soon me irritates
 ‘His eating early irritates me.’

We find a similar situation when it comes to zero-derived deadjectival verbs, (15). The main motivation for postulating a zero morpheme in this context is the existence of overt suffixes with similar functions (cf. the ‘overt analogue criterion’ in Sanders 1988).

- (15) a. L'autore attual-**izz**-a e talvolta riscrive le sue opere.
 The-author updates and sometimes rewrites the his work
 ‘The author updates and sometimes even rewrites his (previous) work.’
- b. Luca attiv- $\emptyset$ -a le notifiche sul cellulare.
 Luca activates the notifications on-the phone.
 ‘Luca activates the notifications on his phone.’

In the example in (15b), assuming the existence of a zero morpheme is again a convenient way of categorising the verbs that are derived differently based on semantic similarities. The suffix *-izz* marks the verb in (15a) as an event, and the zero marking is considered to do the same.

However, many (morpho)syntactic and semantic concerns have been raised regarding the use of zero morphemes (both derivational and inflectional). We briefly go over a few of the most common arguments.

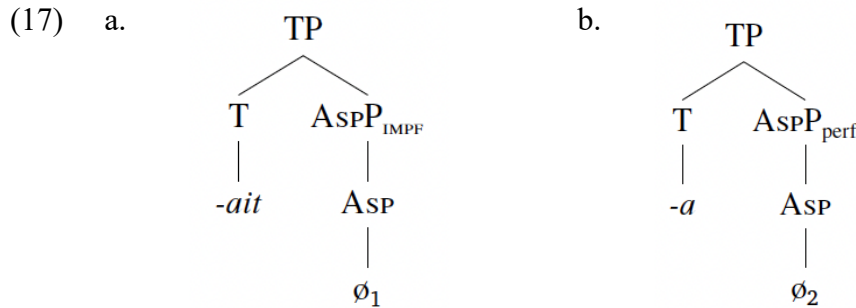
A first argument, brought up by Anderson (1992), has to do with the proliferation of zero morphemes and the question of how they should be ordered with respect to each other or to other overt morphemes. As an example, Anderson mentions the Latin first-person verb form of *amare* ‘love’, which is *am-o*. This verb form must contain information pertaining to

conjugation class, indicative mood, active voice, present tense, and first person and number. However, given the paucity of morphological markers, the majority of these features would have to be realised by zero morphemes, at least in an account that accepts them. Consequently, since most of the information would be encoded silently, it would quickly become unclear what the ordering sequence of features is, and which of those features would have to be expounded by the overt markers *am-* and *-o*, and which ones by a zero. The bracketed structure in (16) show only a small selection of the possible solutions.

- (16) a. [[[[[[[am] $\emptyset_{1\text{CONJ}}$] $\emptyset_{\text{ACTIVE}}$] $\emptyset_{\text{IND}}$] $\emptyset_{\text{PRES}}$] $-\mathbf{0}_1$] $\emptyset_{\text{SG}}$]
b. [[[[[[[am] $\emptyset_1$] $\emptyset_{\text{SG}}$] $\emptyset_{\text{ACTIVE}}$] $\emptyset_{\text{IND}}$] $\emptyset_{1\text{CONJ}}$] $-\mathbf{0}_{\text{PRES}}$]
c. [[[[[[[am] $\emptyset_{\text{IND}}$] $-\mathbf{0}_{\text{SG}}$] $\emptyset_{\text{ACTIVE}}$] $\emptyset_{1\text{CONJ}}$] $\emptyset_1$] $\emptyset_{\text{PRES}}$]

A related issue, discussed in Starke (2014), is the lack of selectional restrictions associated with zero morphemes. He illustrates this problem by means of the French verbal paradigm. In this language, the distinction between perfective and imperfective aspect is (to some extent) preserved in the past tenses. The *imparfait* and *passé simple* form a natural opposition with the former expressing imperfective and the latter perspective aspect. While both tenses have dedicated morphology (e.g. *donn-ait* ‘give-3SG.IMPF’ vs. *donn-a* ‘give-3SG.PAST’), the aspectual contrast is not overly marked. In an account which employs zero morphemes, the existence of two distinct ones would have to be assumed again to realise the difference. However, then, the question arises how each tense selects for the correct zero, (17). As Starke points out, the co-occurrence of the past tense marker *-a* with the perfective zero morpheme, or the co-occurrence of the past tense *-ait* with the imperfective zero morpheme is not

predetermined, it has to be established by additional mechanisms that ensure the right match (e.g. contextual rules).⁸



Finally, regarding zero derivation, Lieber (2004) remarks that plenty of different semantic relations can be discerned between the nominal and adjectival bases and the verbs derived from them (e.g., locative: *to shelve*, resultative: *to dry*, instrumental: *to hammer...* etc.). As Dahl & Fábregas (2018) argue, this multitude of meanings cannot be accounted for by a single zero morpheme, because the zero would have to function as a regular affix, and affixes tend to have

⁸ An anonymous reviewer reasoned that this issue only occurs as a result of the fact that Starke assumes a separate aspectual projection. It could easily be resolved by simply incorporating it under T⁰. However, one argument in favour of an aspectual projection could be the existence of aspectual periphrastic constructions such as *être en train de* ‘be in the process of’. Apart from this, despite the fact that Starke has independent reasons to avoid bundling features under a single, terminal node (as would be the case for T + Asp), his solution is in line with the remark: instead of considering *-ait* or *-a* as the lexicalisation of one TAM head, he argues that they are complex lexical items consisting of multiple syntactic nodes/features which are able to lexicalise the trees in (17). He also explains that this approach has the advantage over one with zero morphology, because it ensures the correct matches between T_{PAST} and IMPF and T_{PAST} and PERF, as these match-ups are lexically defined. As discussed in Sections 4 through 6, we adopt the same approach.

specific meanings. It cannot be explained either by assuming as many zero morphemes as there are meanings, because then one runs into the aforementioned proliferation issue.

Given the issues surrounding zero in general, we propose an alternative analysis in this paper. Specifically, we pursue the idea that zero morphemes can be explained away by means of ‘phrasal lexicalisation’, a tenet of but not exclusive to the Nanosyntactic framework (cf. Starke 2009, Caha et al. 2024 amongst others). To put it in concrete terms, we argue that the VP is internally complex, following Ramchand’s (2008, 2018) well-known decomposition, and that the roots of verbs such as *fermare* ‘stop’ lexicalise bigger parts of this structure than the roots of verbs such as *solidificare* ‘solidify’. We discuss this in more detail in the next section.

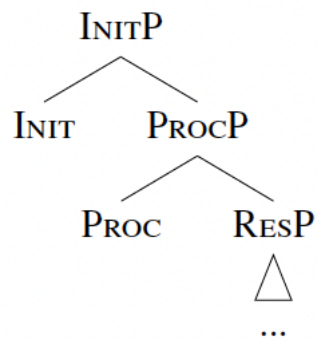
4 Lexicalising deadjectival verbs

In this section, we discuss our main assumptions about the structure of deadjectival verbs (cf. Section 4.1), and provide the basics of Nanosyntax and its model of lexicalisation, applied to said structure (cf. Section 4.2).

4.1 Internal structure of deadjectival verbs

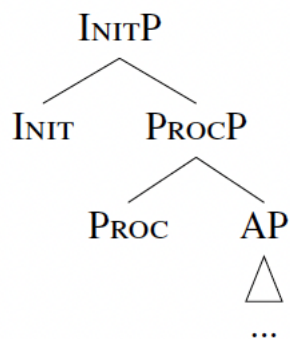
For our analysis, we adopt the underlying verbal structure proposed by Ramchand (2008). Instead of a single, verbal projection, she assumes a decomposition of the different eventive components, resulting in a structure consisting of three heads: Initiation (INIT), Process (PROC) and Result (RES), as shown in (18). Init signals that the event is causative, Proc introduces the actual event, and RES adds the resultant state of the event. A verb may comprise all three heads or a subset of them, depending on what it conveys. That is, each verb minimally needs a PROC head, but INIT and RES are optional and are only present when relevant.

(18)



As for deadjectival verbs, she proposes a similar structure. The main difference is that adjectives are assumed to incorporate a resultative component of their own and therefore replace the RES head in the complement of PROC (cf. contra Vanden Wyngaerd et al. 2022, who, on the basis of Czech, object that not all bare adjectives are resultative).

(19)



The structure in (19) is motivated on morphological and semantic grounds. The containment (i.e., structural embedding) of the adjective within the event is based on the fact that the verbs are often more complex than the adjectives they are derived from and also build on the semantics of those adjectives. This is illustrated by the verbs in (20).

(20)	A	GLOSS	V	GLOSS
	veloc-(e)	‘fast’	veloc-izz-a-re	‘go faster’
	amar-(o)	‘bitter’	amar-eggi-a-re	‘make bitter’
	solid-(o)	‘solid’	solid-ific-a-re	‘solidify’
	stanc-(o)	‘tired’	stanc-a-re	‘tire’

The containment of the inchoative (i.e. PROC) eventive head within the causative (i.e. INIT) one is also reflected by morphological complexity in a number of languages, as is indicated by the examples in (21).⁹

(21) Haspelmath (1993: 91)

- | | | | | |
|----|----------|---------------------|---------|--------|
| a. | Georgian | duγ-s | ‘cook’ | (inch) |
| | | a-duγ-ebs | ‘cook’ | (caus) |
| b. | French | fondre | ‘melt’ | (inch) |
| | | faire fondre | ‘melt’ | (caus) |
| c. | Arabic | darasa | ‘learn’ | (inch) |
| | | darrasa | ‘teach’ | (caus) |

The causative is also argued to entail the inchoative semantically (cf. Causational Entailment in Lundquist et al. 2016). That is, a sentence like (22) is semantically infelicitous because the

⁹ We adopt Haspelmath’s (1993) terminology in this regard. Inchoativity should therefore not be understood to mean ‘indicating the start of an event’ but should be viewed as the opposite of a causative event, that is, an unaccusative or intransitive event or, in more neutral terms, an event without a causer.

causative event ‘Cristina broke the glass’ implicates that an event of breaking has taken place, and it can thus not be negated. In the opposite direction, the same entailment does not hold. The inchoative event ‘The glass broke’ does not necessarily imply that an agent was involved in the event.

- (22) #Cristina ha rotto il bicchiere ma il bicchiere non si
 Cristina has broken the glass but the glass not REFL
 è rotto.
 is broken
 ‘Cristina broke the glass, but the glass did not break.’

Summarising, we assume a decomposed verbal structure in which the adjective is contained within the inchoative, and the inchoative is contained within the causative, as in (19). In the next section, we show how this structure is lexicalised in Nanosyntax.

4.2 Nanosyntax

Starting with the basics, Nanosyntax is a late insertion theory of morphology, which means that syntactic structures are assumed to be merged first and lexicalised afterwards by means of lexical items (in the post-syntactic lexicon). As a consequence, these items are conceived of as links between phonological representations and syntactic trees with abstract features. As an example, consider the English adjective/verb *open*, (23).

- (23) a. The door is **open**.
 b. The door **open**-ed.
 c. Luca **open**-ed the door.

Being able to express a bare adjective, an inchoative event and a causative event, the form *open* is associated to a structure encoding these features, (24). Note that each label between the brackets stands for a node of the form [XP[X]].

(24) /open/ $\Leftrightarrow$ [A PROC INIT]

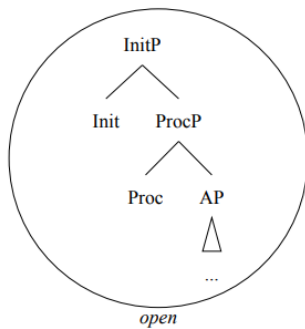
In this view, lexicalisation is essentially a process of matching. Syntax builds structures and searches the lexicon for a lexical item that can realise it. Which items are suitable candidates for lexicalisation and which are not is determined by the Condition on Matching, formulated as in (25).

(25) *Condition on Matching* (Starke 2009)

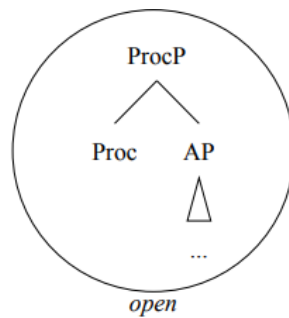
A lexically stored tree matches a syntactic node iff the lexically stored tree contains the syntactic node.

In other words, the condition allows a lexical item to lexicalise a structure if it contains an identical subconstituent (referred to as the Superset Effect), or if it is identical to the full structure. Applied to the lexical item of *open*, it means that *open* could lexicalise all the features it is specified for, but also proper subparts of it, (26). We indicate successful lexicalisations by circles.

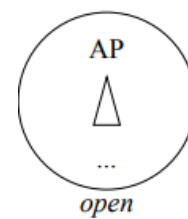
(26) a. Causative



b. Inchoative



c. Adjective



As lexicalisation no longer targets heads but phrasal nodes, and as lexical items like *open* contain and may realise multiple nodes at once, it becomes possible to derive the adjectival, the inchoative and the causative interpretations without the insertion of zero morphemes.

Now, as was mentioned before, not all Italian deadjectival verbs are zero-derived. Similarly, in English, we find adjectives that are verbalised by a suffix, such as *wide-(n)*, (27).

(27) a. The road is **wide**.

b. The road wide-**n**-ed.

c. The workers wide-**n**-ed the road.

As the adjective is not able to express the inchoative nor the causative event without the aid of the suffix *-n*, its lexical item is considerably smaller than that of *open*, (28a). It is the suffix *-n* which is specified for the verbal features instead, (28b).

(28) a. /wide/ ⇔ [A]

b. /n/ ⇔ [PROC INIT]

Given the different specifications of the lexical items in (28), the lexicalisation of the deadjectival structure looks slightly different for *widen* than for *open*.

Nevertheless, the same principles hold: the requirement in the Condition on Matching must be met and only when an identical constituent can be found, can it be mapped onto the corresponding phonological representation. In this case, the adjective is realised by *wide*, but the verbalising features by *-n*. We have illustrated this so far by means of tree structures, but we can also visualise this in a less technical format. In Nanosyntax, it is also common to summarise derivations in so-called lexicalisation tables, as the one in (29), that keep track of which morphemes lexicalise which features (from left to right).

(29)

	A	PROC	INIT
‘wide.CAUS’	wide	n	

In the next section, we illustrate the lexicalisation process further by providing a walkthrough of the derivations of zero-derived and suffixed verbs in Italian. Just as in the English examples in this section, the distinction between the two types of Italian verbs essentially boils down to a distinction in the ‘size’ of the lexical items, i.e., the items store more or fewer features. Because of this, some require an additional marker, whereas others do not.

5 Analysis: ‘Zero’-derivation vs. suffixation

In this section, we go over the derivations of zero-derived verbs (cf. Section 5.1) and suffixed verbs (cf. Section 5.2). As the gist of the analysis comes close to what we presented more

generally for English in the previous section, we suggest that readers who are familiar with Nanosyntax proceed immediately to Section 6.

Now, before we move on to the derivations, we must briefly say something more about the specific lexicalisation procedure. While we have already established that syntactic structures undergo lexicalisation through late insertion, and that only lexical items which are identical to a (sub)constituent of the structure can generate a match (cf. Condition on Matching in (25)), we have not mentioned anything yet about how the structures are constructed or what happens when a match cannot be (immediately) found.

In Nanosyntax, syntactic structures are assumed to be built incrementally and lexicalised cyclically. That is, features are merged one by one, and after each merge, the current structure is checked against the lexicon for an available item to lexicalise it.

If lexicalisation is successful, the derivation continues, a new feature is added to the structure and the cycle is renewed (i.e., syntax checks the lexicon but now for a match of $[F[G]]$ instead of just F , and this whole process repeats itself). However, if there is no lexical item which corresponds to the current structure, syntax must perform a number of evacuation movements (also referred to as lexicalisation-driven movements) to remedy the situation. These movements are described in and ordered by the so-called lexicalisation algorithm, (30), a step-by-step manual which syntax follows rigidly and systematically.¹⁰

¹⁰ Various updated versions of the algorithm have been proposed to capture a wider range of phenomena (cf. Starke 2024, Caha et al. 2024, Starke 2022, De Clercq 2020 amongst others). We stick to the original formulation in Starke (2018), as it keeps things relatively simple, while still serving the purposes of this paper and staying true to the essence of the most recent version. Additionally, the other versions of the algorithm provide the same outcomes.

(30) *Lexicalisation algorithm* (Starke 2018)

- a. Merge F and lexicalise FP
- b. If (a) fails, move the spec of the complement of F, and retry (a)
- c. If (b) fails, move the complement of F, and lexicalise FP
- d. If (c) fails, go back to the previous cycle and try the next option for that cycle

According to the algorithm, a feature F is merged first and tried for lexicalisation, (a). If that fails, there are three repair operations: spec-movement, (b), complement movement, (c), or backtracking to the previous cycle, (d). After each repair operation, lexicalisation of FP is attempted again. Important to note is that these movements do not leave traces.

With this in mind, we proceed with a more detailed discussion of how the two types of verbs are derived without the use of zero morphemes. The algorithm will be illustrated throughout.

5.1 'Zero'-marked verbs

The verb that we will use as a representative of the whole class of zero-derived A-verbs is *fermare* 'stop'. We repeat the relevant data in (31).¹¹

¹¹ We choose *fermare* for the sake of presenting a clear picture. Unlike some other verbs, *fermare* allows for an inchoative reading without the reflexive marker *si*; it is optional. As our discussion of *si* in Section 2.2 indicated,

- (31) a. Lucia è una persona ferma, seria e saggia.
 Lucia is a person firm serious and wise
 ‘Lucia is a firm, serious and wise person.’
- b. Il treno (si) ferma in ogni stazione.
 The train (REFL) stops in every station
 ‘The train stops at every station.’
- c. Il controllore ferma il treno.
 The conductor stops the train
 ‘The conductor stops the train.’

Given the fact that the root *ferm-* creates adjectives, inchoatives, and causatives, we argue it is associated to the syntactic structure in (32).

(32) /*ferm*/ ⇔ [A PROC INIT]

The structure in (32) indicates that the root, over the course of the derivation, will be able to lexicalise the adjectival as well as the verbal features without the intervention of an additional, verbalising marker. Since all the features are contained within a single morpheme, and since Nanosyntax adopts a type of cumulative exponence, no zeroes need to be postulated.

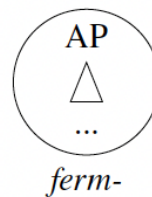
though, it is improbable that the presence of the reflexive marker would considerably alter the core argument of this paper.

To derive all three interpretations, syntax starts by building the adjective, (33a). The lexical item *ferm-* is a suitable candidate to lexicalise it due to the Condition on Matching. If we recall, the natural consequence of matching based on identity is that lexical items can also lexicalise subconstituents that are contained in their structure, (33b).

(33) a. Merge FP

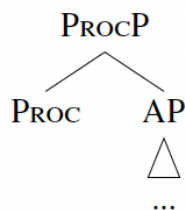


b. Lexicalise FP

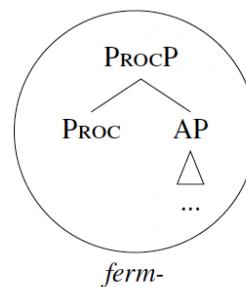


As the lexicalisation of AP is successful, syntax proceeds to merge the next feature in the sequence, i.e., PROC, and consults the lexicon for a new matching item, (34a). The resulting structure is again contained within the lexical item of *ferm-*, so the inchoative reading is obtained with the same item, (34b). Note that any previous lexicalisations are overridden.

(34) a. Merge F



b. Lexicalise FP



The derivation of zero-derived A-verbs is thus relatively straightforward. By encompassing all the features, the roots form a match for the full deadjectival verb structure and its

subconstituents.¹² We visualise this with the table in (35). The dark grey cells show how the root lexicalises every feature at every cycle.

(35)

	A	PROC	INIT
‘firm’	firm		
‘stop.INCH’	firm		
‘stop.CAUS’	firm		

5.2 Suffixed verbs

We now turn to the derivation of suffixed verbs. For this, we use the verb *solidificare* ‘solidify’. We repeat the relevant data in (37). As we can see, the adjective *solid-a* is distinguished from the derived verbs by means of the verbalising affix *-ific*. Leaving it out yields ungrammatical sentences.¹³

¹² To prevent other zero-derived verbs to override *ferm-* at any given stage of the derivation, Caha et al. (2019, 2021) argue for a Faithfulness restriction, which, in informal terms, states that free choice applies at the start of each derivation, but that conceptual information is not overridden from that point onwards. That is, if the root *ferm-* lexicalises AP and then gets verbalised, it is not expected to be replaced by a root that is semantically unrelated, such as *attiv-* ‘activ(e)/ate’.

¹³ A reviewer observes that an analysis without zeroes would lead to two predictions that do not hold, at least in the case of Spanish. First, such an analysis would predict that overtly derived nouns or adjectives should not be able to become verbs by simply adding the theme vowel. However, in Spanish, this word formation process appears to be quite productive (e.g., *limitación* > *limitacionar* ‘limit’, *nervioso* > *nerviosar* ‘make nervous’). In contrast, to the best of our knowledge, Italian *-zion(e)* nouns are rarely converted into verbs (e.g., *attenzione* > *attenzionare* ‘pay attention to’, but *limitazione* > **limitazionare*), and such formations are generally confined to a bureaucratic register (cf. Bongi 2009). The same pattern is observed with deadjectival bases, which are primarily verbalized through parasynthesis (e.g., *nervoso* > *innervosire* ‘make nervous’). Therefore, it appears that this issue

- (37) a. Ci vuole fiducia per un'amicizia solida.
 It wants trust for a-friendship solid
 'You need trust to have a solid friendship.'
- b. La vernice (si) solidifica/*solida.
 The paint (REFL) solidifies
 'The paint solidifies.'
- c. La vernice protettiva solidifica/*solida le tue creazioni.
 The paint protective solidifies the your creations
 'The protective paint solidifies your creations.'

It thus seems that the root *solid-* contains fewer features than *ferm-* because it can only express the adjective on its own. As such, it is associated to the adjectival features only, as shown in (38a). The remaining features are linked to the suffix *-ific*, as shown in (38b).

- (38) a. /*solid*/ ⇔ [A]

is primarily relevant to Spanish, rather than to Italian, which is the focus of this paper. We leave this issue for future research, though.

Second, an analysis without zeroes would predict that a verb should not have both a suffixed and a non-suffixed variant. In section 2.1, we briefly mentioned the existence of some rare doublets, such as *attivare* / *attivizzare* ('activate') and *massimare* / *massimizzare* ('maximize'). De Clercq & Vanden Wyngaerd (2019) propose an analysis based on pointers to account for this type of allomorphy, originally developed to explain the idiomatic meanings of syntactic constituents. While we cannot go into the details of this mechanism here, we refer the reader to De Clercq & Vanden Wyngaerd (2019) for a full explanation of how this can be accounted for.

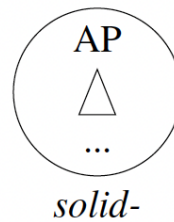
- b. $/if\acute{c}/ \Leftrightarrow [\text{PROC INIT}]$

The derivation of this verb then starts out in the same way as before. Syntax first merges the adjective, (39a), and searches a fitting lexical item. As *solid-* is the only item which corresponds to AP, there is a perfect match and lexicalisation is successful, (39b).

- (39) a. Merge F

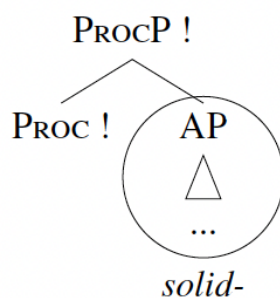


- b. Lexicalise F



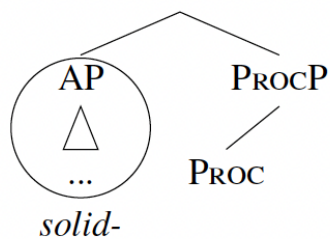
At the next step, where PROC is merged however, (40a), the derivation diverges slightly from what we have seen so far. There are no lexical items which form a match with the newly built structure (as is indicated by the '!'), so syntax must try an evacuation movement (cf. steps (b) and (c) in the algorithm in (30)). The first one is spec-movement, but this is not possible here, as there is no specifier of AP yet, (40b). The second one is complement movement and this works. The sister of PROC and all the features contained within this phrase are pied-piped across PROC, obtaining a structure that can be matched, (40c). AP is still lexicalised by *solid-* and PROC is now a separate constituent that is contained within the lexical item *-if\acute{c}*, (40d).

- (40) a. Merge F

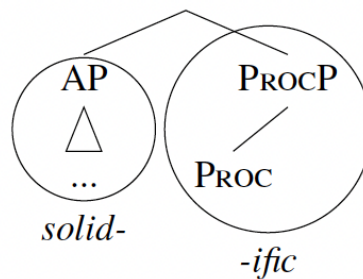


- b. Lexicalise FP

c. Move complement

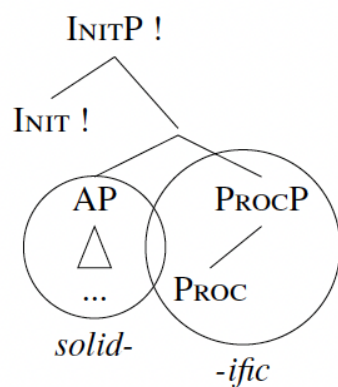


d. Lexicalise FP

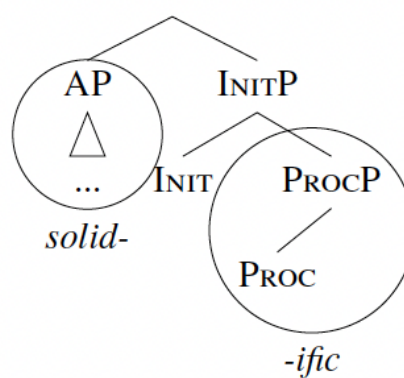


As the final step, syntax then merges INIT. Again, we do not have a stored lexical item that matches the new structure, (41a). However, this time syntax is able to perform spec-movement; it is possible to evacuate the AP node which forms the specifier of PROC, (41b), and move it across. As a result, INIT now forms a constituent with PROC, and syntax finds a matching lexical item in *-ific* again, (41c).

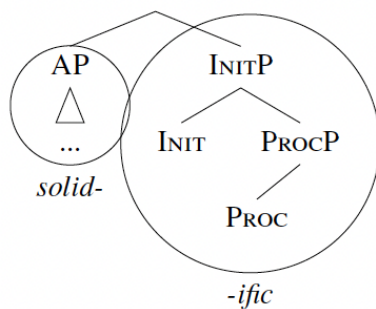
(41) a. Merge F



b. Move spec



c. Lexicalise FP



The difference between zero-derived and suffixed verbs can thus be reduced to a difference in the size of the lexical items instead of a difference between zero or overt exponence. In our view, zero-derived verbs, as *fermare*, have roots that are bigger / associated to more features, whereas suffixed verbs, as *solidificare*, are smaller / associated to fewer features. In order to verbalise, the latter therefore require a dedicated marker that lexicalises PROC and INIT, the former does not. This provides a more constrained explanation for the distribution of suffixes. We provide a comparative overview in the table in (42). The dark grey cells again indicate the roots, the lighter ones the verbaliser.

(42)

	A	PROC	INIT
‘stop.CAUS’	ferm		
‘solidify.CAUS’	solid	ific	

6 Beyond the verbal core

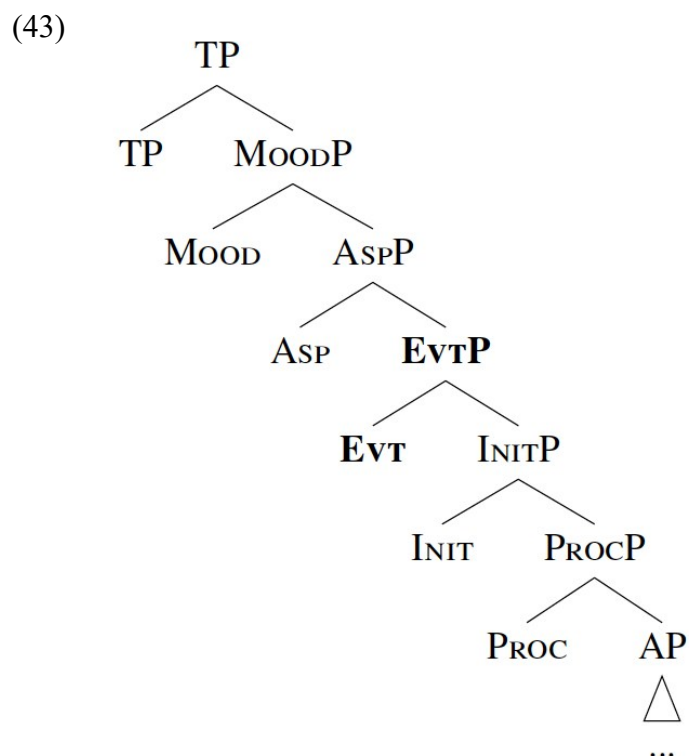
In this last section, we go beyond the core of the verbal structure and show that zero morphemes in the inflectional domain can be eliminated as well with the same technology, i.e., phrasal lexicalisation. We start with a short discussion of the theme vowel *-a* in Section 6.1 and have a

look at the TAM and agreement markings in Section 6.2. For now, we limit ourselves to the present tense paradigms.

6.1 Theme vowel *a*

As suggested by Fábregas (2022) for Spanish derived verbs, we assume that the theme vowel *-a* is the overt realisation of a syntactic head that ensures the transformation of a lexical event into an eventuality, i.e., an event with time and world parameters.

More concretely, Fábregas proposes that this syntactic head corresponds to Ramchand's (2018) Event head (EVT). This head does not add any descriptive content to the eventuality in terms of argument structure or Aktionsart, but it enables the lexical verb to combine with grammatical aspect, mood, and tense, so that it can project as a clause in the syntactic spine. As such, it is situated in between INIT and ASP, (43).



If the theme vowel is the exponent of EVT, it allows us to explain why verbs are marked by a theme vowel in the first place. Without a realisation of the Event layer, the verb cannot become a fully-fledged event in the Davidsonian sense of the word (cf. Fábregas 2022).

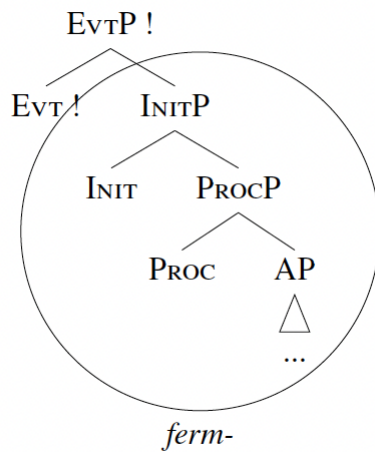
In this conception, the theme vowel thus has a clearly delineated syntactic and semantic role (as opposed to what is assumed in approaches which treat theme vowels as ornamental morphology, e.g., Oltra-Massuet 1999).

Regarding the exact implementation of this in our account, we will assume (for now) that *-a* is associated with a single Event feature, as in (44).

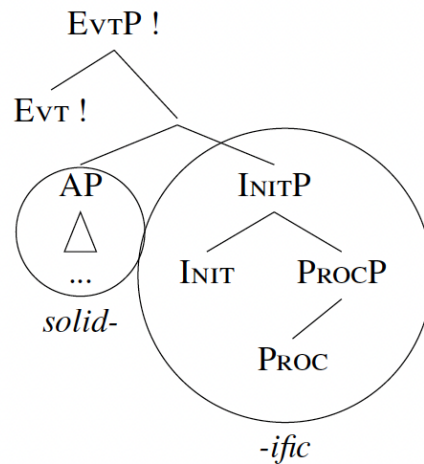
(44) */a/* ⇔ [EVT]

As a consequence, when EVT is merged on top of the VP structures of both zero-derived, (45a), and suffixed verbs, (45b), there are no immediate matches in either situation, and syntax must perform evacuation movements. In the case of *fermare*, there is no other option than to move the complement, (45c). In the case of *solidificare*, spec-movement is attempted, but does not lead to the lexicalisation of EVT, as there is no lexical item containing [EVT INIT PROC]. Here too, complement movement yields the desired outcome, (45d).

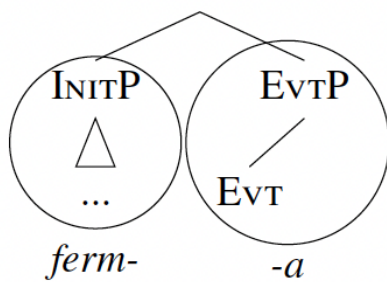
(45) a. Merge F



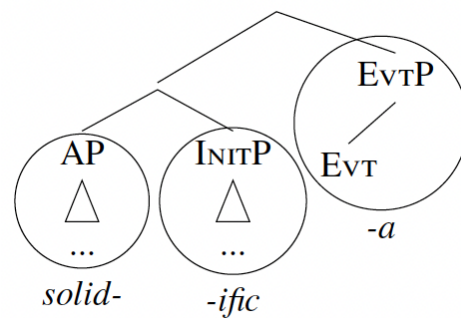
b. Merge F



c. Move complement



d. Move complement



The addition of Evt thus does not lead to any surprises yet; there is one additional head and one corresponding lexical item to realise it, as summarised in (46).¹⁴

¹⁴ Note that in languages that do not have theme vowels (e.g. English) or that have athematic verbs such as French (e.g., *partir* ‘to leave’, *vendre* ‘to sell’, cf. Pomino & Remberger 2022) there would be a potential zero morpheme as well as a potential lack of an Event head due to the lack of overt morphology. Such cases will have to be explained away by assuming that the roots incorporate Evt in their lexical structure, or if present, the verbalising suffix. This is not so strange given that many syncretic instances can be found where the root would be expected to lexicalise minimally [PROC EVT ASP T] and sometimes even $\varnothing$, e.g. *I open, you open, we open, they open* etc.

(46)

	A	PROC	INIT	EVT
‘stop.CAUS’	firm			a
‘solidify.CAUS’	solid	ific		a

6.2 Tam and agreement markings

We now move on to the remaining inflectional parts of the verbal structure. In the table in (47), we present the present tense paradigms of *firmare* and *solidificare*. We focus on the endings indicated in bold.¹⁵

(47)

	<i>firmare</i> ‘stop’	<i>solidificare</i> ‘solidify’
1SG	firm- o	solidific- o
2SG	firm- i	solidific h-i
3SG	firm- a	solidific- a
1PL	firm-ia- mo	solidific h-ia-mo
2PL	firm-a- te	solidific-a- te
3PL	firm-a- no	solidific h-a-no

¹⁵ A note on two particularities concerning the paradigm in (47). First, note that the *-ch* which appears in 2SG and 1PL of the verb *solidificare* is not a phonological alternant, but simply an orthographic requirement to indicate that the pronunciation of the last consonant remains [k] throughout. This is necessary since a subsequent front vowel otherwise leads to the palatalisation of the plosive [k], e.g. *vinci* [vintʃi] ‘you win’. Second, the theme element *-ia* in the 1PL goes back to the Latin theme of present subjunctive, *-ea(mus)* or *-ia(mus)* respectively. From a phonetic point of view, already in Vulgar Latin, the unstressed short vowels E and I in a hiatus developed into the glide /j/ which then spread to all conjugations in Old Florentine, eventually also reaching the indicative form (cf. Vincent 1980).

If we recall the discussion on zero morphology in Section 3, one of the main concerns about zeroes was that there are very few theoretical restrictions on the distribution and amount a system can or should allow, or on the competition with overt lexical items. Under this view, we could, for instance, assume that in the 1SG *ferm-o* ‘I stop’, the suffix *-o* is the first-person marker (in analogy to other forms of the full verbal paradigm that show the same overt marker, such as *ferm-a-v-o* ‘I stopped’). If this is the case, however, we are left many other unrealised features, such as conjugation, number, tense, aspect, and mood, and each of them could be lexicalised by a (distinct) zero morpheme, (48a), (cf. Anderson’s (1992) Latin example). The same issue would arise for the 2SG, (48b).

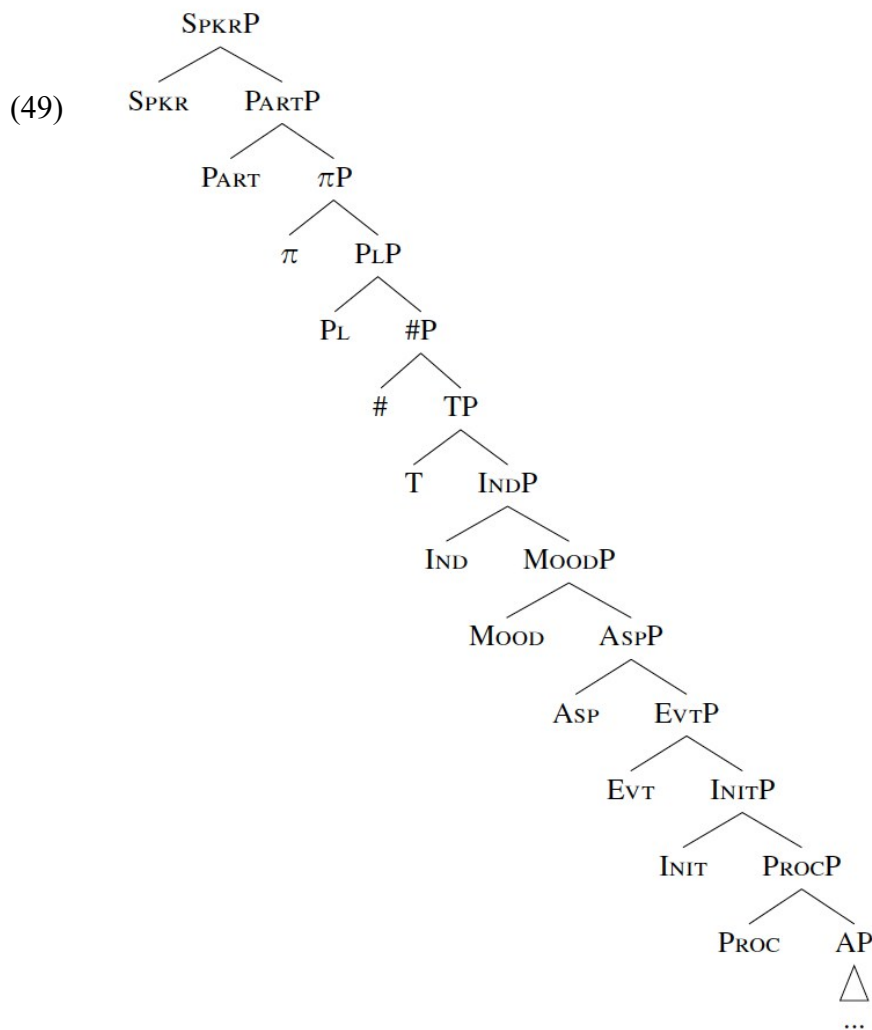
- (48) a. [[[[[ferm ø_{1CONJ}] ø_{ASPECT}] ø_{MOOD}] ø_T] -**o**₁] ø_{SG}
 b. [[[[[ferm ø_{1CONJ}] ø_{ASPECT}] ø_{MOOD}] ø_T] -**i**₂] ø_{SG}

An equally problematic case is the 3SG form. While all the forms in (47) seem to exhibit suffixes that could be interpreted as inflectional endings, the 3SG does not. To explain this, some accounts are compelled to introduce even more zero morphemes, either because they assume that the theme vowel is missing and the exponent *-a* simply becomes the 3sg marker (e.g., *ferm- $\emptyset$ -a* in Maiden 2018), or because they hypothesise that the exponent of 3SG is a zero (e.g., *ferm-a- $\emptyset$* in Pomino & Remberger 2019).

Evidently, neither are desirable outcomes. If the 3sg has a zero exponent for EVT, it would require us to postulate additional rules or mechanisms to explain the competition between *-a* and zero in this specific context. If *-a* is the exponent of EVT and everything else is zero, we run into the proliferation issue once more.

However, if we assume a uniform structure and adopt the same Nanosyntactic model of lexicalisation (including phrasal lexicalisation as well as the lexicalisation algorithm), as we did in the previous sections, we can circumvent the challenges that zero morphology poses and derive the whole paradigm in a more straightforward way. All it takes is varying the sizes of the lexical items involved. In what follows, we briefly discuss the extended functional sequence used in recent work on verbal systems across Romance (cf. Starke 2020 for French, Starke & Cortiula 2021 for Brazilian Portuguese, Fábregas 2022 for Spanish, Cortiula 2023 for Friulan) and show the derivations of all verb forms.

The functional sequence we adopt is provided in (49), for the 1PL form. This form maximises the quantity of number (#) and person (π) features.



At the bottom of the structure, we find the core layers [INIT PROC A] that constitute deadjectival verbal roots (cf. Section 4). Following those are the features for aspect (ASP), mood (MOOD), indicative (IND) and tense (T).

Regarding mood, empirical evidence suggests that the indicative is more complex than the subjunctive, on a semantic, syntactic and morphological level. As Cortiula (2023) discusses, indicatives have their own time reference and are able to stand alone in root clauses, (50a), whereas subjunctives are time-dependent and occur in embedded clauses, (50b).¹⁶ In addition, Starke (2020, 2021) and Starke & Cortiula (2021) argued that several morphological alternations in the French and Brazilian verb paradigms can be resolved if the indicative indeed contains more features than the subjunctive.

- (50) a. Vado al supermercato domani.
 go.1SG to-the supermarket tomorrow
 ‘I am going to the supermarket tomorrow.’
- b. Spero che vada al supermercato domani.
 hope.1SG that go.3SG to-the supermarket tomorrow
 ‘I hope that s/he goes to the supermarket tomorrow.’

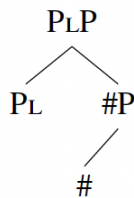
With respect to tense, we assume that T stands for a default present tense. This is uncontroversial, as the present tense is viewed as the default or unmarked tense in most

¹⁶ Note that subjunctives can also appear in main clauses - particularly when expressing desire, exhortation, or supposition. In these cases, even though the subjunctive is used in a main clause, it still behaves similarly to a subordinate clause due to the elliptical structure, where the main clause is implied but not explicitly stated (e.g., *Che tu possa arrivare sano e salvo!* ‘May you arrive safe and sound!’) (cf. Giorgi 2009).

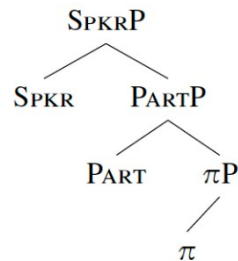
Romance languages (e.g., Oltra-Massuet 1999), compared to the past or future tense (e.g., *ferma-v-o* ‘I stopped’ and *ferm-er-ò* ‘I will stop’). In the case of more complex marked forms, such as the past tense, syntax will merge an additional, corresponding feature.

In the highest areas of the tree we find the agreement features, comprising number and person (for the motivation of the Person features, cf. Harbour 2016, Ackema & Neeleman 2018, Vanden Wyngaerd 2018 amongst others, for the motivation of their particular ordering, cf. Starke 2020). # is the default number feature, expressing, on its own, a singular form, and plural forms (PL) are built from it. Third person forms only contain the π node. Second person forms encode both π and an additional participant (PART) feature. First person expressions are tripartite with an additional speaker feature (SPKR) added to π and PART. These features are thus assumed to be structurally contained within each other, as shown in (51a) and (51b).

(51a) a. Number



b. Person



With the verbal structure in place, let us now go through the derivations of the TAM and agreement markings, starting with the singular forms.

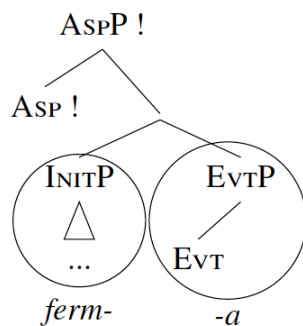
Previously in Section 5.2, we proposed that the theme vowel *-a* is best analysed as a functional head EVT that ensures the equipment of the lexical event with world and time specifications. However, since we learned that *-a* is the only exponent available to express

TAM and agreement in 3SG, we need to revise the lexical item and associate it to more features if we wish to avoid introducing zero suffixes. The updated lexical item is provided in (52). As we can see, it is now specified for Aspect, default Mood, Indicative, default present Tense, default singular number and default third person, all in addition to Event. This is to account for the fact that we get the 3SG forms *ferm-a* and *solidific-a*.

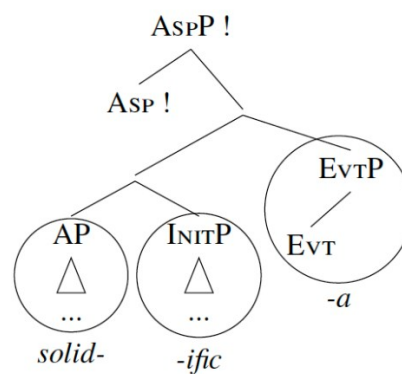
$$(52) \quad /a/ \Leftrightarrow [\text{EVT} \quad \text{ASP} \quad \text{MOOD} \quad \text{IND} \quad \text{T} \quad \# \quad \pi]$$

To derive the full 3rd person structure, we start again from the situation in (45a) and (45b). Syntax merged EVT for the two types of verbs and lexicalised it by executing complement movements in both cases. As the next step, syntax will merge ASP, (53a-b), and in order to lexicalise it by *-a*, the specifiers in each structure have to move so that the new feature can form a constituent with EVT, (53c-d).

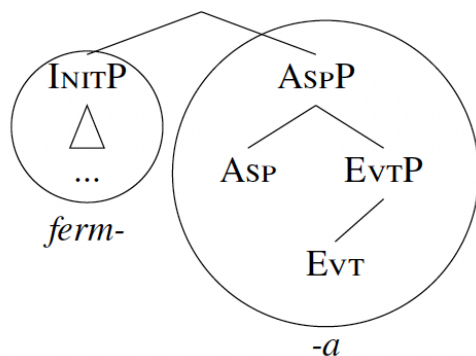
(53) a. Merge F



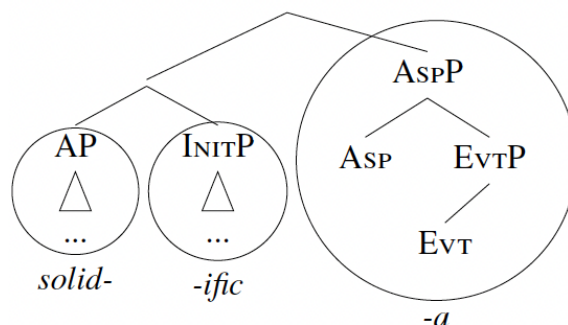
b. Merge F



c. Move specifier



d. Move specifier



Syntax will keep adding the other features and perform a spec-movement in each cycle in order to lexicalise all the features of 3SG by *-a*. The result is shown in the table in (54).

(54)

	A	PROC	INIT	EVT	ASP	MOOD	IND	T	#	π
'stop.CAUS'	ferm			a						
'solidify.CAUS'	solid	ific			a					

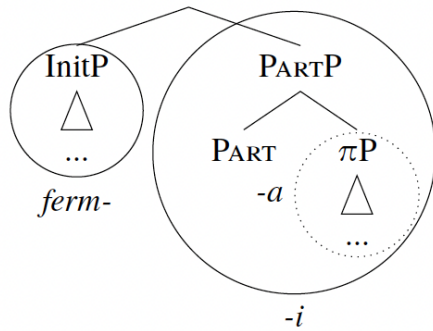
The other lexical items *-i* and *-o* for 2SG and 1SG, minimally have to contain the dedicated person features PART and SPKR to distinguish them from 3SG and each other. Their exact specifications, though, depend on the stance one adopts towards the status of the theme vowel. On the one hand, one could opt for a purely morphological solution and assume that *-a* is completely overridden by *-i* and *-o*.¹⁷ In that case, the lowermost feature of both lexical items would be EVT, just like for A, to maintain constituency, (55).

¹⁷ We wish to emphasise here that a complete override would not constitute a violation of the Faithfulness Condition mentioned in footnote 9. The suffixes are purely functional lexical items and are not assumed to have conceptual information.

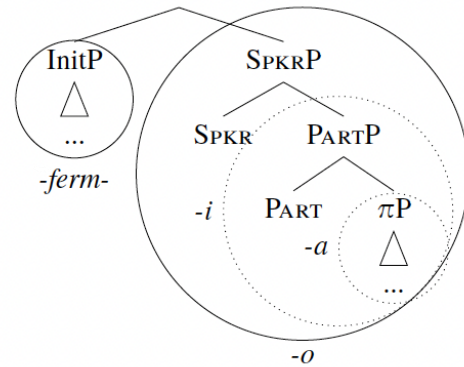
- (55) a. $/i/ \Leftrightarrow [\text{EVT} \quad \text{ASP} \quad \text{MOOD} \quad \text{IND} \quad \text{T} \quad \# \quad \pi \quad \text{PART}]$
- b. $/o/ \Leftrightarrow [\text{EVT} \quad \text{ASP} \quad \text{MOOD} \quad \text{IND} \quad \text{T} \quad \# \quad \pi \quad \text{PART} \quad \text{SPKR}]$

A derivational advantage to this option is that no spatial technology would be required. When PART is merged and made to form a constituent with π , for instance, the lexical item *-i* simply becomes the better match than *-a*. Similarly, when SPKR is added, the lexical item *-o* becomes the only suitable candidate, as is shown in (56a-b). A disadvantage is that one loses uniformity and would have to assume that there is no theme vowel in 2SG and 1SG (in line with for instance Maiden 2018 who argues for this with respect to 3SG).

(56) a. Override



b. Override



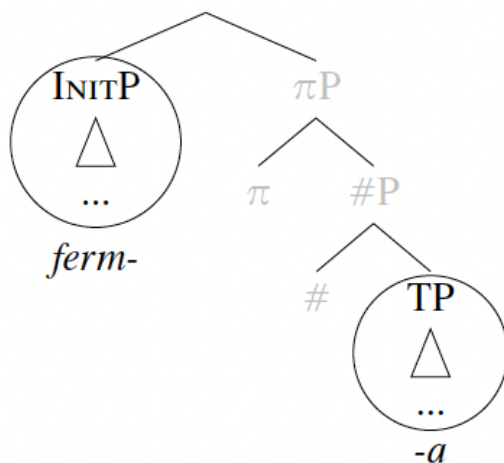
On the other hand, one could also decide that the disappearance of *-a* is mainly a phonological effect (cf. vowel deletion in the context of another vowel, e.g. Napoli & Vogel 1990), but that it is preserved in the underlying structure. In that case, the lowermost feature of *-i* and *-o* could go as low as Aspect. However, pending evidence to the contrary, it is best to assume the smallest set of features necessary and to opt for $\#$ as their lowermost feature to indicate that they are singular markers, (57).

(57) a. $/i/ \Leftrightarrow [\# \ \pi \ \text{PART}]$

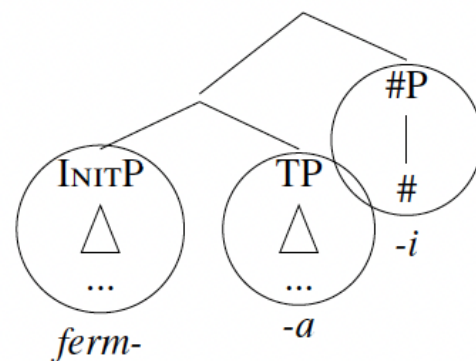
b. $/o/ \Leftrightarrow [\# \ \pi \ \text{PART} \ \text{SPKR}]$

The advantage of this approach is that a certain parallel can be maintained with the 3sg and the plural forms, and that the idea is preserved that the theme vowel is responsible for the connection between the lexical root and the time and world parameters. The disadvantage is that it adds to the technical load, because the derivation will require backtracking, as shown in (58): When PART is merged, and syntax does not find a matching item, even after exhausting all the rescue movements, it will have to backtrack and try something else, (58a). Instead of lexicalising # with *-a*, complement movement will take place and *-I* will be inserted instead, (58b), as it is the next step in the algorithm syntax can try. When the structure is eventually sent to PF, *-a* will be deleted.

(58) a. Backtracking



b. Moving complement



Regardless of which option one decides upon, the singular paradigm can be derived without the introduction of zeroes. We summarise the forms in (59).

(59)

		A	PROC	INIT	EVT	ASP	MOOD	IND	T	#	π	PART	SPKR
'stop.CAUS'	1		ferm				(a)					o	
	2		ferm				(a)				i		
	3		ferm				a						
'solidify.CAUS'	1	solid	ific				(a)					o	
	2		ific				(a)				i		
	3		ific				a						

Turning now to the plural, we start by repeating the forms in (60), for the sake of convenience.

(60)

	<i>fermare</i> 'stop'	<i>solidificare</i> 'solidify'
1 PL	ferm-ia- mo	solidifich-ia- mo
2 PL	ferm-a- te	solidific-a- te
3 PL	ferm-a- no	solidifich-a- no

As we can see, there are three dedicated morphemes, *-mo*, *-te* and *-no*, all preceded by the theme vowel. We therefore assume that their lexical items look as follows, (61). They contain features related to number and person. The feature that sets them apart from the singular suffixes is PL, and what sets them apart from each other are the different person features.

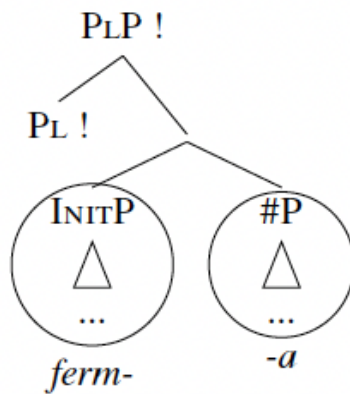
(61) a. $/mo/ \Leftrightarrow [\pi \text{ PL PART SPKR}]$

b. $/te/ \Leftrightarrow [\pi \text{ PL PART}]$

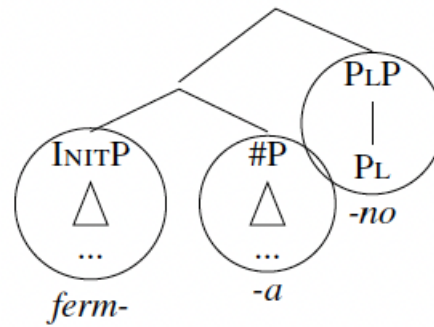
c. $/no/ \Leftrightarrow [\pi \text{ PL}]$

The derivation of these forms is less complicated than that of the singular forms. At the point when PL is introduced in the structure, (62a), syntax needs to perform a complement movement to isolate the plural feature, so that it may be lexicalised by *-no* for the third person plural, (62b).

(62) a. Merge F



b. Moving complement



When the remaining person features are added, syntax is able to lexicalise all of them through various spec-movements of the unlabelled node that dominates INITP and #P. The final stages of the plural derivation are then presented in (64).

(64)

		A	PROC	INIT	EVT	ASP	MOOD	IND	T	#	PL	π	PART	SPKR
'stop.CAUS'	1		ferm				ia					-mo		
	2		ferm				a					-te		
	3		ferm				a				-no			
'solidify.CAUS'	1	solid	ific				ia					-mo		
	2	solid	ific				a					-te		
	3	solid	ific				a				-no			

To sum up, in this section we extended the discussion concerning the necessity of positing zero morphemes from the core verbal domain to the inflectional verbal components. We provided an analysis that does not rely on zero morphology by assuming that TAM and agreement features – just like the core verbal features – are pronounced cumulatively by overt morphemes.

7 Conclusion

The main objective of this paper was to show that there is no need for zero morphology in the derivation of Italian zero-derived and suffixed deadjectival A-verbs (contra traditional approaches, e.g. Scalise 1984, Thornton 1990 amongst others). Specifically, we proposed a formal account where zeroes are eliminated as an effect of cumulative exponence or, in this case, phrasal lexicalisation.

Based on a fine-grained syntactic decomposition of the verbal structure, we argued that although there are two types of derived verbs, they consist of the same underlying features but differ in the number of features lexicalised by the root. Zero-derived verbs have roots that lexicalise all the core verbal features, while suffixed verbs have roots that can only lexicalise a small portion of the features and need an additional suffix that takes care of the remaining features. In a similar fashion, we also accounted for the present tense indicative forms. We eradicated the need for zero morphemes in the exponence of TAM and agreement features by varying the size of the various lexical items involved in the conjugation of *fermare* and *solidificare* (and by extension other verbs belonging to either categories) and allowing them to lexicalise different parts of a stable verbal functional sequence.

The analysis we presented in this paper is still deficient in certain areas, however. As discussed in Section 2, Italian has several verbalisers, i.e., many allomorphic variants. Although they seem to convey very similar meanings for the most part, there are verb pairs where the semantic contribution of the suffix may play a bigger role than initially assumed. For instance, to account for the difference between *toscanizzare* ‘use Tuscan (phonology/spelling)’ and *toscaneggiare* ‘to act as a Tuscan’, we would have to reconsider the featural configuration of the single suffixes (see Fábregas 2022). We hope to return to this issue and tackle allomorphy between the various suffixes in the future.

Moreover, as briefly touched upon in Section 2, it would be worthwhile to examine verbs where the inchoative appears obligatorily with the reflexive marker *si*. We mentioned that there are good reasons to assume that *si* is linked to the aspectual layer of the structure. However, we will need to inspect the bigger picture since deadjectival verbs do not exist in a ‘vacuum’, i.e., they should be analysed along with the large group of anticausative verbs.

Finally, as mentioned in Section 6, we would have to introduce additional features in order to account for other tenses. In the imperfect tense, for instance, we would also need to evaluate the overt morphemes and their corresponding functions. Recall the verb *fermare* ‘stop’ in the 3PL *ferm-a-v-a-no*. We would have to deal with an additional morpheme *-a* between the exponent of T (*-v*) and the agreement morpheme *-no* that looks like the theme vowel that is following the root. However, in the 1sg *ferm-a-v-o* this morpheme is absent, which recalls the absence of the theme vowel in the 1SG and 2SG of the present tense indicative (*ferm-o*). In that case, we proposed that *-o* also incorporates EVT. We could assume the same for the imperfect forms. However, we would still need to account for the presence of two theme vowels. We also leave this as an avenue for future research.

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